VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi-590 014

A Major Project Phase-1 Report

On

“MULTIPLE DISEASE PREDICTION USING MACHINE LEARNING ALGORITHMS”

Submitted in partial fulfillment of the requirements for the

PROJECT PHASE-1 (BCD685) course of the 6th semester

Bachelor of Engineering

In

Computer Science & Engineering (DATA SCIENCE)

Submitted by

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CERTIFICATE

This is to certify that the Major Project phase-1 work entitled **"MULTIPLE DISEASE PREDICTION USING MACHINE LEARNING ALGORITHMS"** is a bonafied work carried out by **Ms. Amrutha B V(4AI22CD003), Ms. Ananya B K (4AI22CD004), Ms. Bhumika K(4AI22CD009), Ms. Hitha B R (4AI22CD026)** in partial fulfillment for the **Major Project (BCS685)** course of 6th semester Bachelor of Engineering in **Computer Science and Engineering (Data Science)** of the Visvesvaraya Technological University, Belagavi during the academic year **2024-2025**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The Major project report has been approved as it satisfies the academic requirements in respect of Project Work prescribed for the said Degree.

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ABSTRACT PHASE-1

The project introduces an advanced Health Assistant Application built using Python, aimed at predicting multiple diseases through integrated machine learning models. This application features four specialized prediction models-Diabetes, Heart Disease, Parkinson's Disease and Breast Cancer—each designed to analyze user-provided health parameters. By entering the required clinical metrics, users can obtain automated predictions generated by pre- trained algorithms tailored to each condition. The interface incorporates a streamlined layout, enabling seamless navigation across different prediction sections. Every module includes clearly defined input fields, ensuring that users provide accurate data essential for reliable outcomes. Once the information is submitted, the system processes the inputs and delivers a prediction indicating the probability of the respective disease, accompanied by graphical visualizations that illustrate the input ranges. The use of machine learning in healthcare demonstrates how technology can support early prediction.

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Chapter 1

Introduction

1.1 Background

The integration of technology in healthcare has revolutionized the way medical diagnoses are conducted. With the advent of machine learning (ML), healthcare professionals can leverage data-driven insights to enhance patient care. Disease prediction systems have emerged as vital tools, enabling early prediction and intervention, which can significantly improve patient outcomes. This project focuses on developing a Health Assistant application that utilizes machine learning models to predict various diseases, including Diabetes, Heart Disease, Parkinson's Disease and Breast Cancer. By providing users with immediate health insights, the application aims to empower individuals to take proactive steps towards their health.

1.2 Motivation

- Chronic diseases like Diabetes, Heart disease, Parkinson’s and Breast cancer are major global health concerns.
- It is crucial for effective treatment and improved survival rates.
- Traditional diagnostic methods are often time-consuming, costly and not easily accessible.
- Many individuals lack awareness of disease symptoms and risk factors.
- Machine learning offers the potential for fast, accurate and automated health predictions.
- A web-based application can provide an accessible, user-friendly platform to promote proactive health management and awareness.
- Early detection of chronic diseases is often delayed due to limited access to timely, affordable, and reliable diagnostic methods.
- Machine learning combined with a web-based platform can provide quick, accessible and user-friendly health predictions, promoting awareness and proactive health management.

1.3 Problem Statement

Prediction of diseases like Diabetes, Heart Disease, Parkinson’s and Breast Cancer is often delayed due to limited access to timely diagnosis. There is a need for a simple, accessible tool that can provide quick health predictions. This project develops a web application using machine learning models to predict these diseases from user-provided health parameters, enabling proactive health management and awareness.

1.4 Objective of the System

1.4.1 Predictive Model Development

The project focuses on developing and training machine learning models for four major diseases: Diabetes, Heart Disease, Parkinson's Disease, and Breast Cancer. Each model is trained using relevant datasets to ensure accurate and reliable disease prediction.

1.4.2 User-Friendly Interface

A simple and intuitive web interface is designed, allowing users to easily input their health parameters. The structured design ensures smooth navigation and accessibility for all users.

1.4.3 Accuracy and Reliability

The system emphasizes accuracy by utilizing pre-trained and validated models to deliver timely predictions. This ensures that users receive trustworthy insights into their health conditions.

1.4.4 User Education

Along with predictions, the application educates users about common symptoms and risk factors. This feature helps in spreading awareness and encourages healthier lifestyle choices.

1.4.5 Early Diagnosis and Awareness

By combining predictions with awareness, the application promotes early diagnosis of critical diseases. This proactive approach empowers users to take timely action, leading to better health management and outcomes.

1.4.6 Data Preprocessing and Model Efficiency

Focus on preparing and cleaning health datasets to improve the performance and reliability of machine learning models.

1.4.7 Proactive Health Management

Provide instant health insights that motivate users to seek timely medical advice and adopt preventive measures.

1.5 Scope of the Project

- Development of a web application that integrates multiple disease prediction modules.
- Implementation of machine learning algorithms along with necessary techniques.

- Focus on four specific diseases: Diabetes, Heart Disease, Parkinson’s Disease and Breast Cancer.
- Utilization of pre-trained models to generate predictions based on user-provided health parameters.
- User interface design to ensure easy navigation and smooth interaction for users.
- Clarifying that the system is not a replacement for medical advice, while suggesting directions for future improvements.
- The system supports educational awareness by informing users about symptoms and risk factors.
- The project explores future improvements such as integration with wearable devices, multilingual support and cloud-based deployment for scalability.

Chapter 2

Literature Survey

Title 1: "Multiple Disease Prediction System using Machine Learning"

Description:

The paper presents a Multiple Disease Prediction System (MDPS) using machine learning to predict various illnesses simultaneously. It addresses healthcare inefficiencies caused by disease-specific models, enabling early detection, accurate diagnosis, and personalized treatment. Emphasizing scalability, security, and real-world application, MDPS advances precision medicine and improves patient outcomes.

Remarks:

- **Integration of ML in Healthcare** – The paper effectively emphasizes how machine learning can transform fragmented healthcare into a unified predictive model.
- **Focus on Practical Utility** – It highlights not only technical advancements but also usability, scalability, and integration into real-world clinical practices.
- **Addressing Key Challenges** – Ethical issues, patient data security, and the adaptability of the system are well-recognized, making the approach comprehensive and forward-looking.

Title 2: "Disease Prediction: Smart Disease Prediction System using Random Forest Algorithm"

Description:

The paper presents a Smart Disease Prediction System using the Random Forest algorithm, trained on 4920 patient records with 12 symptoms and 41 diseases. Achieving 95% accuracy, the system aids in early diagnosis, reducing healthcare delays. It emphasizes practical implementation, GUI design, and real-world utility in medical decision support.

Remarks:

- **Strong Accuracy** – Achieves 95% classification accuracy, proving the reliability of Random Forest for disease prediction.
- **Real-World Utility** – GUI-based system enables doctors and patients to predict diseases easily from symptoms.
- **Scalability** – Designed to handle multiple diseases simultaneously, making it suitable for practical healthcare applications.

Title 3: " Heart Disease Prediction using Hybrid machine Learning Model"

Description:

The paper "Heart Disease Prediction using Hybrid Machine Learning Model" focuses on predicting cardiovascular diseases using the Cleveland Heart Disease dataset. It applies Random Forest, Decision Tree, and a Hybrid Model (combination of both) for classification. The hybrid model achieves around 88.7% accuracy, showing better results than individual methods. A block diagram illustrates the workflow from user input to predicted output, highlighting the role of machine learning in early and accurate heart disease detection.

Remarks:

- **Accuracy Enhancement** – Hybrid models improve accuracy and reliability compared to single algorithms.
- **Life-Saving Impact** – Early detection with ML can save lives and reduce treatment costs.
- **Practical Implementation** – Real-world hospital use requires further testing and validation.

Title 4: "A Machine Learning Approach to Predictive Modeling for Breast Cancer Prediction"

Description:

The paper "A Machine Learning Approach to Predictive Modeling for Breast Cancer Prediction" emphasizes the importance of accurate diagnostic methods for early breast cancer detection. Using the UCI Breast Cancer dataset, various algorithms such as Naïve Bayes, Random Forest, Decision Trees, KNN, Logistic Regression, and SVM were applied, achieving accuracy levels between 94% and 97%. The study also explores ensemble methods, cross-validation, and feature selection strategies to enhance model performance, aiming to support medical practitioners in timely and reliable breast cancer diagnosis.

Remarks:

- **High Accuracy & Reliability** – Multiple ML algorithms show strong accuracy, with ensemble methods further improving results.
- **Survival & Treatment Impact** – Early detection through predictive models can significantly improve patient survival and treatment planning.
- **Future Applications** – The research highlights potential for medical use, but real-world adoption still requires clinical validation.

Title 5: "Diabetes Disease Prediction using Machine Learning on Big Data of Healthcare"

Description:

The paper "Diabetes Disease Prediction using Machine Learning on Big Data of Healthcare" addresses the growing challenge of predicting diabetes using large-scale healthcare data. By applying algorithms such as Naïve Bayes, Support Vector Machine (SVM), Random Forest, and Simple CART through the WEKA tool, the study evaluates their performance. Results show that SVM delivers the highest accuracy for diabetes prediction. The research highlights the role of Big Data Analytics and ML in improving healthcare outcomes by enabling automated diagnosis and supporting clinical decision-making.

Remarks:

- **Best Performing Algorithm** – Among tested algorithms, SVM achieved the highest accuracy for diabetes prediction.
- **Insights & Early Diagnosis** – Big Data with ML provides valuable insights for timely detection and better healthcare planning.
- **Implementation Challenges** – Real-world use requires tackling issues like data quality, scalability, and clinical validation.

Chapter 3

Tools and Technology Specification

3.1 Software Requirement

Backend

- **Programming Language –** Implemented in Python 3.8 or higher.
- **ML Libraries –** TensorFlow, Scikit-Learn, NumPy, and Pandas.
- **Detection Logic –** Core engine responsible for image/data processing and prediction.
- **Database Support –** Stores and manages user-uploaded images/data securely.
- **API Services –** Provides endpoints for client-server communication.
- **Data Processing –** Performs preprocessing and inference using trained ML models.
- **Cloud Hosting –** Deployable on AWS, Google Cloud, or Azure platforms.
- **Security Features –** Ensures encrypted communication, authentication, and controlled access.
- **Scalability –** Capable of supporting multiple users and large-scale processing.
- **Monitoring –** Records logs, tracks system performance, and analyzes model behavior.

Frontend

- **Operating System** – Supports Windows 10/11, Linux, and macOS.
- **Web Browser** – Compatible with Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **UI Access** – Accessible through any modern web browser without installation.
- **Framework** – Built using Flask to support interactive and responsive UI.
- **Cross-Platform** – Responsive interface compatible with desktop, laptop, and mobile devices.
- **Lightweight Scripts** – Utilizes minimal Python scripts for fast UI rendering and performance.
- **Secure Communication** – Enables safe client–server data transfer to prevent data leakage.
- **Data Upload** – Allows users to upload images or datasets for analysis and prediction.
- **Real-Time Display** – Shows processing results instantly after model execution.
- **Visualization** – Provides graphical representation of model output and processed data.

3.2 Hardware Requirement

3.2.1 For End Users (Client-Side):

- **Processor:** Intel Core i3 7th Generation or above
- **Ram:** Minimum 4GB or higher
- **Storage:** minimum 2gb of free disk space
- **Display:** minimum 1024x768 screen resolution
- **Internet connection:** Required for real-time data processing and model update

3.2.2 For Server (If Hosted on a Cloud or Local Machine):

- **Processor:** Intel Xeon / AMD Ryzen 7 or higher for high-performance computation
- **Ram:** minimum 16gb (recommended: 32gb for handling large-scale requests)
- **Storage:** minimum 100gb SSD for faster read/write operations
- **Graphics card (optional):** nvidia gpu for deep learning model acceleration (if required)
- **Operating system:** Windows Server, Linux (Ubuntu/CentOS), or cloud hosting (AWS, Azure, Google Cloud)

Chapter 4

System Design

The system design of the Multiple Disease Prediction model starts with the User Input stage, where the user provides necessary disease symptoms and other relevant health parameters. This data then goes through the Data Processing phase, where it is cleaned, normalized and transformed into a suitable format for machine learning. Once processed, the data is split into training and testing datasets to ensure that the model learns effectively while also being tested for accuracy. This structured flow ensures that the input data is accurate, reliable and ready for model training.

In the ML Model Training stage, various algorithms such as Support Vector Machine (SVM), Random Forest, and Logistic Regression are applied to train the model using the training dataset. Each algorithm learns unique patterns from the medical data to predict diseases accurately. The trained model is then evaluated using the testing dataset to check its performance metrics like accuracy, precision, recall, and F1 score. Finally, the system generates a Predicted Output, which is displayed in the User Application, allowing users to receive instant and understandable health predictions for multiple diseases.

4.1 System Architecture

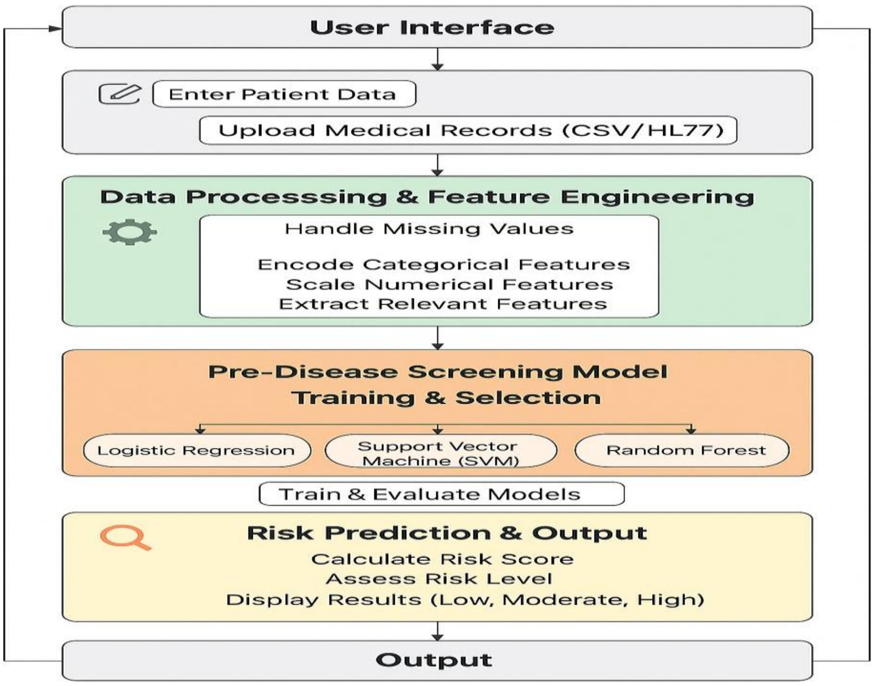

Fig 4.1: System Architecture

The Fig 4.1 illustrates the complete workflow of a system architecture of the Multiple Disease Prediction model is designed to efficiently handle user data and provide accurate health predictions. It starts with users entering their health details through a simple interface. The input data is then processed to remove noise, normalize values and prepare it for training. Machine learning algorithms such as Support Vector Machine, Random Forest and Logistic Regression analyze the data to predict diseases. Finally, the system generates an output that is displayed in an easy-to-understand format, ensuring smooth user experience and reliable prediction accuracy.

Explanation:

1. User Interaction:

- Users provide medical data through a simple and interactive interface.
- The system ensures smooth and fast communication between user and application.

2. Input Handling:

- Collects important health parameters like age, glucose, and blood pressure.
- Validates user inputs to avoid missing or incorrect health information.

3. Data Preprocessing:

- Cleans and normalizes raw data for better algorithm performance.
- Removes duplicates, fills missing values, and selects relevant features.

4. Output Processor:

- Displays disease prediction results in a clear and understandable format.
- Provides accurate feedback with precision, recall, and accuracy scores.

5. Algorithms:

- Uses SVM, Random Forest, and Logistic Regression for predictions.
- Selects the best-performing model based on evaluation metrics.

6. System Integration:

- Connects all modules to ensure continuous data flow between stages.
- Integrates preprocessing, training, and prediction components into one seamless system.

4.2 Use case Diagram

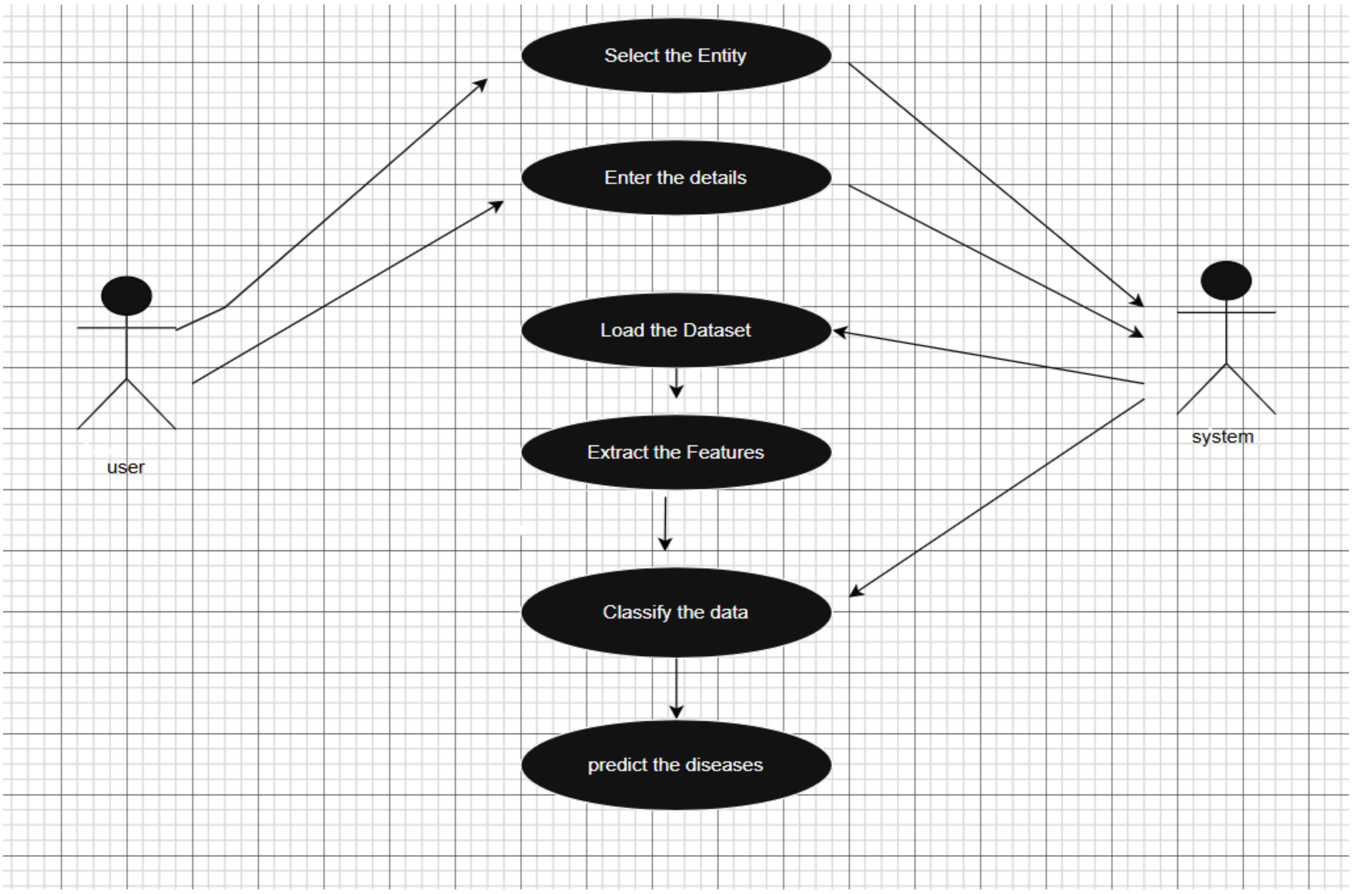

Fig 4.2: Use case Diagram

The figure 4.2 illustrates the Use Case Diagram visually represents the interaction between the user and the system for disease prediction. It defines the main actions users perform and how the system responds to each process, from data input to prediction output.

1. Select the Entity:

The user selects the disease type or entity, determining which prediction model and dataset will be used for processing.

2. Enter the Details:

The user inputs personal or medical details such as age, blood pressure, and glucose levels required for prediction accuracy.

3. Load the Dataset:

The system loads a predefined medical dataset containing patient records to train and test the machine learning models effectively.

4. Extract the Features:

Important parameters or features are extracted from the dataset to identify relevant attributes for disease classification and prediction accuracy.

5. Classify the Data:

The system applies machine learning algorithms to categorize the input data into specific classes like diabetic or non-diabetic results.

6. Predict the Diseases:

Based on the trained model, the system generates final prediction results indicating whether the patient has a disease or not.

7. User:

The user interacts with the system by entering inputs, selecting entities, and viewing disease prediction results through a simple interface.

8. System:

The backend system processes all data-related operations, executes algorithms, and produces prediction outputs for user interpretation and medical insight.

4.3 Data Flow Diagram

The data flow begins with user inputs, where the user provides medical and symptom-related information through the system interface. This input data is forwarded to the analysis module, where machine learning algorithms process and evaluate the data. Based on the selected disease category, the data is routed to the corresponding prediction models such as diabetes, heart disease, Parkinson's, or breast cancer models. Each model analyzes the input independently and generates prediction results. Finally, the system consolidates the outputs and presents the final disease prediction along with recommendations to the user.

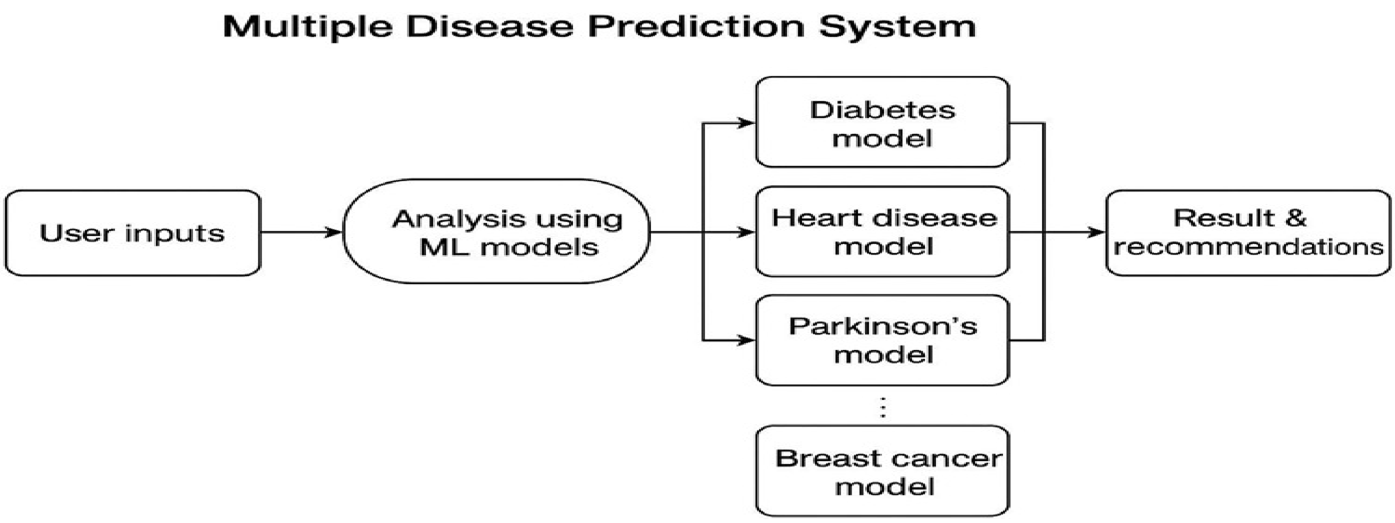

Fig 4.3: Data Flow Diagram

The Data Flow Diagram (DFD) of the Multiple Disease Prediction System provides a clear representation of how data moves through various stages of the system to generate accurate health predictions. The process begins when the user inputs medical details or other parameters through the interface. This raw data is then sent to the data processing module, where it undergoes cleaning, normalization and transformation to ensure consistency and remove any missing or irrelevant values.

Next, the dataset is divided into training and testing subsets, allowing the machine learning models to learn from one portion and validate their performance on another. The machine learning model training stage involves using algorithms like Support Vector Machine (SVM), Random Forest, Logistic Regression and others to analyze complex relationships between input features and disease types. Once trained, these models can accurately classify data and predict multiple diseases such as diabetes, heart disease, Parkinson's or Breast Cancer.

The prediction output is then generated and passed to the user application, where it is displayed in an understandable format. The system may also visualize the results using graphs or accuracy metrics like Precision, Recall, Accuracy, and F1-score. This data flow ensures an efficient, automated process where user inputs are transformed into meaningful, data-driven health insights, supporting early disease detection and decision-making.

Chapter 5

Conclusion

The **Health Assistant** is an innovative web app that utilizes pre-trained machine learning models to predict the likelihood of various diseases such as Diabetes, Heart disease, Parkinson's and Breast cancer. It offers an intuitive and user-friendly interface where users can easily input their health parameters to receive instant, data-driven predictions. The system aims to promote early prediction and awareness empowering individuals to take preventive actions and make informed health decisions. While it provides quick and reliable preliminary assessments, it is not a replacement for professional medical consultation or diagnosis.

Future advancements could significantly enhance its capabilities by integrating additional disease prediction modules, improving accuracy through larger and more diverse datasets and incorporating real-time health monitoring. The integration of personalized health recommendations, data visualization dashboards, and cloud storage for health records can further transform it into a comprehensive digital health companion. Such developments would make the Health Assistant a more powerful and accessible tool for healthcare and wellness management.

REFERENCE

[1] T. Raj, D. Singh and A. Kaur, **"A Machine Learning-Based Framework for Prediction of Weather-Related Diseases,"** International Journal of Advanced Computer Science and Applications (IJACSA), vol. 11, no. 5, pp. 123–130, 2020.

[2] M. Pawar and A. S. Kapse, **"Disease Prediction Using Machine Learning Algorithms,"** International Conference on Intelligent Computing and Control Systems (ICICCS), pp. 461–466, 2021. doi:10.1109/ICICCS51141.2021.9432164.

[3] A. Joshi and S. Kulkarni, **"Health Monitoring and Prediction System Using Machine Learning,"** International Research Journal of Engineering and Technology (IRJET), vol. 8, no. 6, pp. 1500–1505, 2021.

[4] **OpenWeatherAPI Documentation,** OpenWeather Ltd., Available at: https://openweathermap.org/api

[5] Orvile, **"Weather-Related Disease Prediction Dataset,"** Kaggle, Available at: https://www.kaggle.com/datasets/orvile/weather-related-disease-prediction-dataset

[6] J. Brownlee, **"Data Pre-Processing Techniques for Machine Learning,"** Machine Learning Mastery, 2019. Available at: https://machinelearningmastery.com

[7] S. Raschka and V. Mirjalili, **"Python Machine Learning,"** 3rd edition, Packt Publishing, 2019.

[8] T. Hastie, R. Tibshirani, and J. Friedman, **"The Elements of Statistical Learning: Data Mining, Inference, and Prediction,"** Springer, 2017.